Date: 20/10/2026 **SAI DATTA BHAAVAN MBNP**
241401087 B.Tech CSBS

ZOHO CREATOR API INTEGRATION

AIM

To integrate an external REST API into a Zoho Creator application by writing a custom Deluge function that uses the invokeUrl task, and to call that function from a form workflow so that the Loan Application form automatically verifies the applicant's location from the PIN code entered.

PROCEDURE

1. Background

An API (Application Programming Interface) lets one software system request data or services from another over the internet. Most modern web APIs are REST APIs: the client sends an HTTP request (such as GET) to a URL, and the server replies with data, usually in JSON format. Zoho Creator can take part in such integrations in two directions:

- Inbound – other systems call Creator through its own REST API (secured with OAuth 2.0) or through Custom APIs.
- Outbound – Creator calls an external service from inside Deluge using the invokeUrl task.

The Custom API option was not available in the trial account used for this lab, so the outbound approach was implemented. It is also the more useful one for the Loan Application app, because it lets the application enrich its own records with data from a public service.

2. Integration Design

The Loan Application form already stores the applicant's residential address. Two new fields were added – pincode and verified_location. When a record is submitted, a workflow passes the PIN code to a custom function. The function sends a GET request to the India Post PIN code API and extracts the district and state from the reply. The workflow then stores that text in verified_location. A mismatch between the typed address and the verified location can then be spotted by the loan officer.

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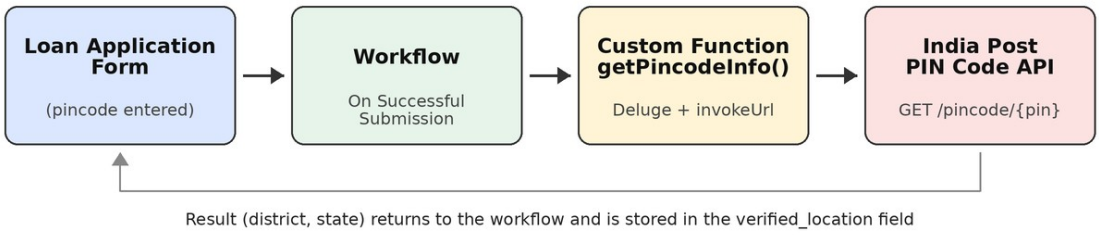


Fig. 1: Data flow of the API integration

3. API Used

Property	Details
Service	India Post PIN code lookup (public, free)
Endpoint	https://api.postalpincode.in/pincode/{pin}
HTTP method	GET
Authentication	None required
Response format	JSON list; first item contains Status and a PostOffice list
Fields used	Status, PostOffice[0].District, PostOffice[0].State

4. New Fields in the Form

Field Name	Type	Purpose
pincode	Single Line	Applicant enters the 6-digit PIN code
verified_location	Single Line	Filled automatically by the workflow with “District, State”; left blank by the user

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5. Creating the Custom Function

- Open the application in edit mode and go to Workflow → Functions → New Function.
- Choose “Execute a function when called”. Name it getPincodeInfo, language Deluge, namespace Default, return type string.
- Add one argument named pin of type string, then click Create Function.
- Replace the template with the code below and click Save.

```
string getPincodeInfo(string pin)
{
    response = invokeUrl
    [
        url :"https://api.postalpincode.in/pincode/" + pin
        type :GET
    ];
    info response;
    result = "Not Found";
    if(response.get(0).get("Status") == "Success")
    {
        office = response.get(0).get("PostOffice").get(0);
        result = office.get("District") + ", " + office.get("State");
    }
    return result;
}
```

The function builds the URL by appending the PIN code, calls the API with invokeUrl, logs the raw response with info, and returns the district and state. If the API reports anything other than Success, it returns “Not Found”.

6. Testing the Function

Before using the function in a workflow, it was run with the Execute button. The test PIN code 600001 was entered as the pin argument. The function executed successfully and the log showed the raw JSON returned by the API (Status: Success, 7 post offices found).

7. Creating the Workflow

- Go to Workflow → Form Workflows → Create New.
- Form: Loan_Application_Form. Run when a record is Created. Trigger: Successful form submission.
- Name the workflow Verify Location On Submission and click Create Workflow.
- Add a Deluge script action with the code below and save.

```
if(pincode != null && pincode != "")
{
    location = thisapp.getPincodeInfo(pincode.trim());
    verified_location = location;
}
else
{
    verified_location = "Pincode not provided";
}
```

The workflow reads the pincode field, calls the custom function through thisapp, and assigns the returned text to verified_location.

8. Testing the Application

The application was opened in access mode and new records were submitted with different PIN codes while leaving verified_location empty. The records were then opened from the report to confirm that the field had been filled by the workflow.

PIN entered	Expected location	Value stored
600001	Chennai, Tamil Nadu	Chennai, Tamil Nadu
560001	Bangalore, Karnataka	Bangalore, Karnataka

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OUTPUT

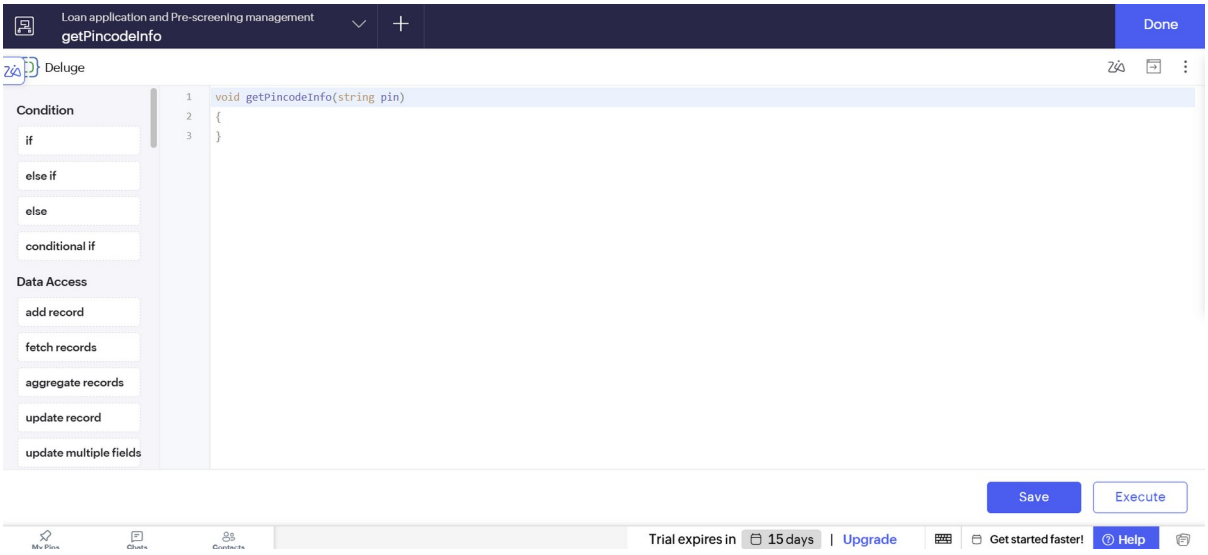


Fig. 2: Custom function getPincodeInfo created with the string argument pin

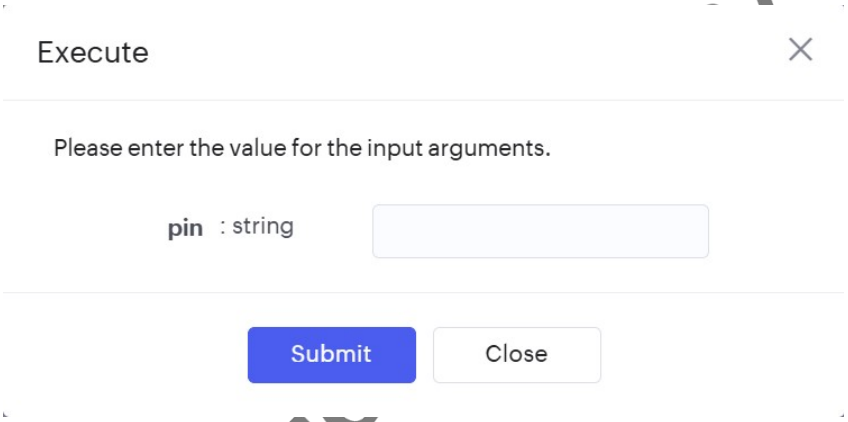


Fig. 3: Executing the function – entering the test PIN code for the pin argument

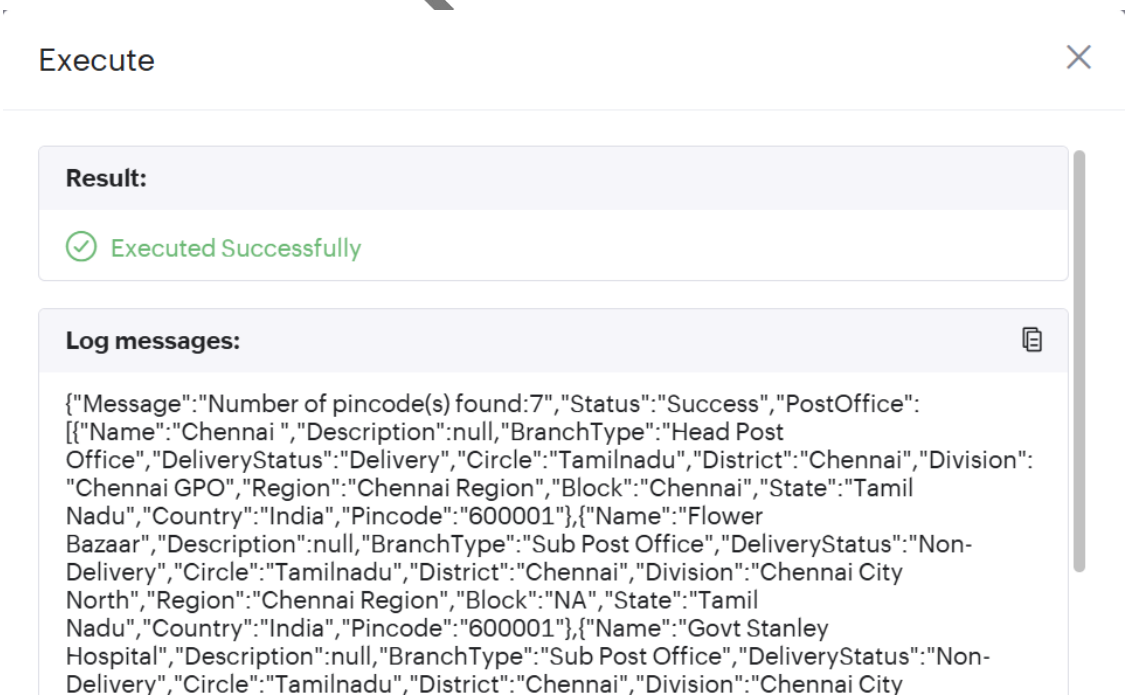


Fig. 4: Function executed successfully – log shows the JSON response from the India Post API

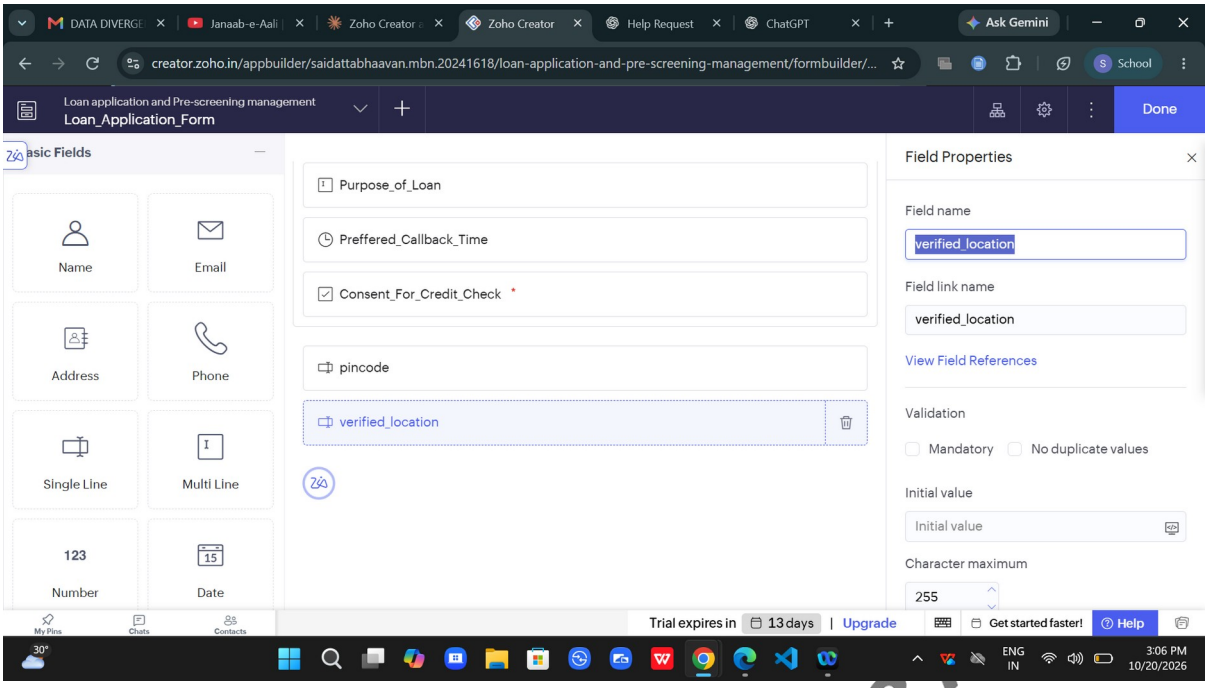


Fig. 5: Loan_Application_Form with the new pincode and verified_location fields

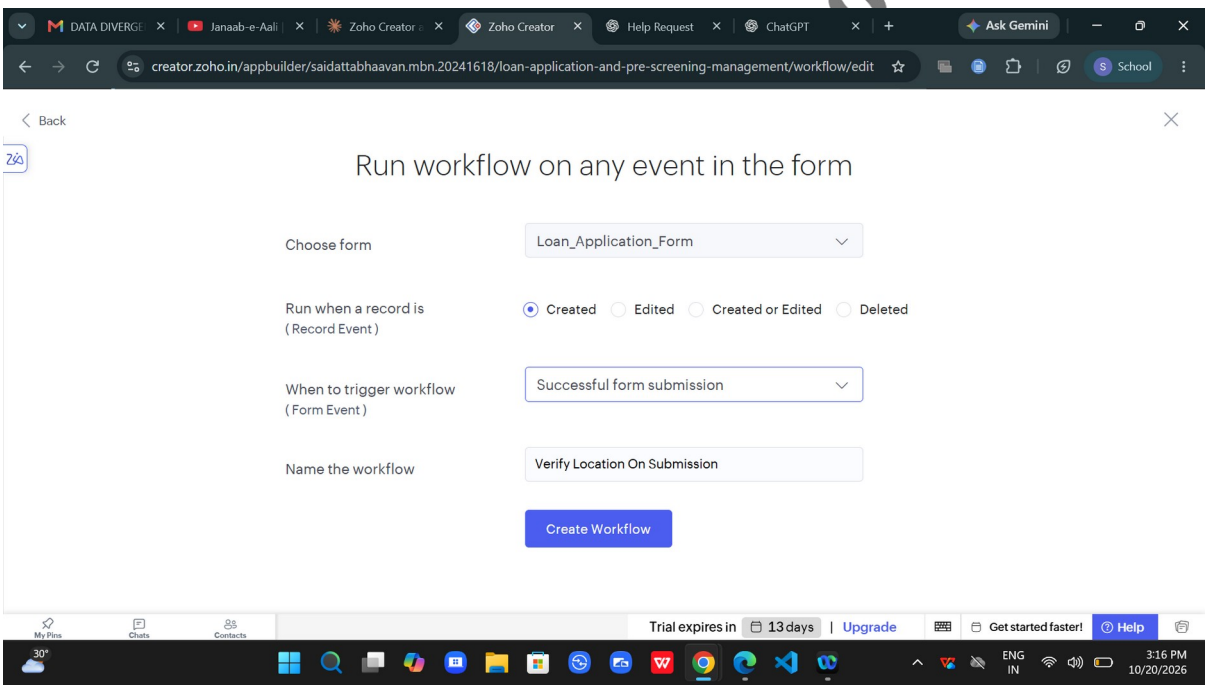


Fig. 6: Creating the form workflow "Verify Location On Submission" (Created → Successful form submission)

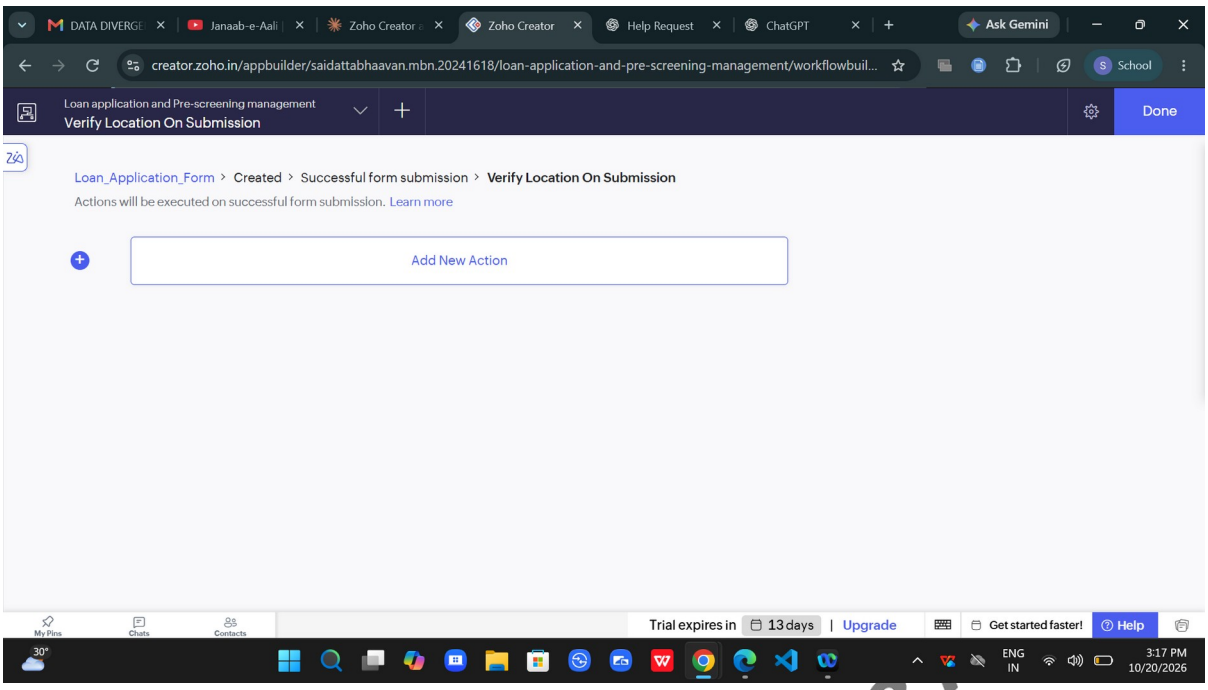


Fig. 7: Workflow builder – the Deluge action is added here to call getPincodeInfo

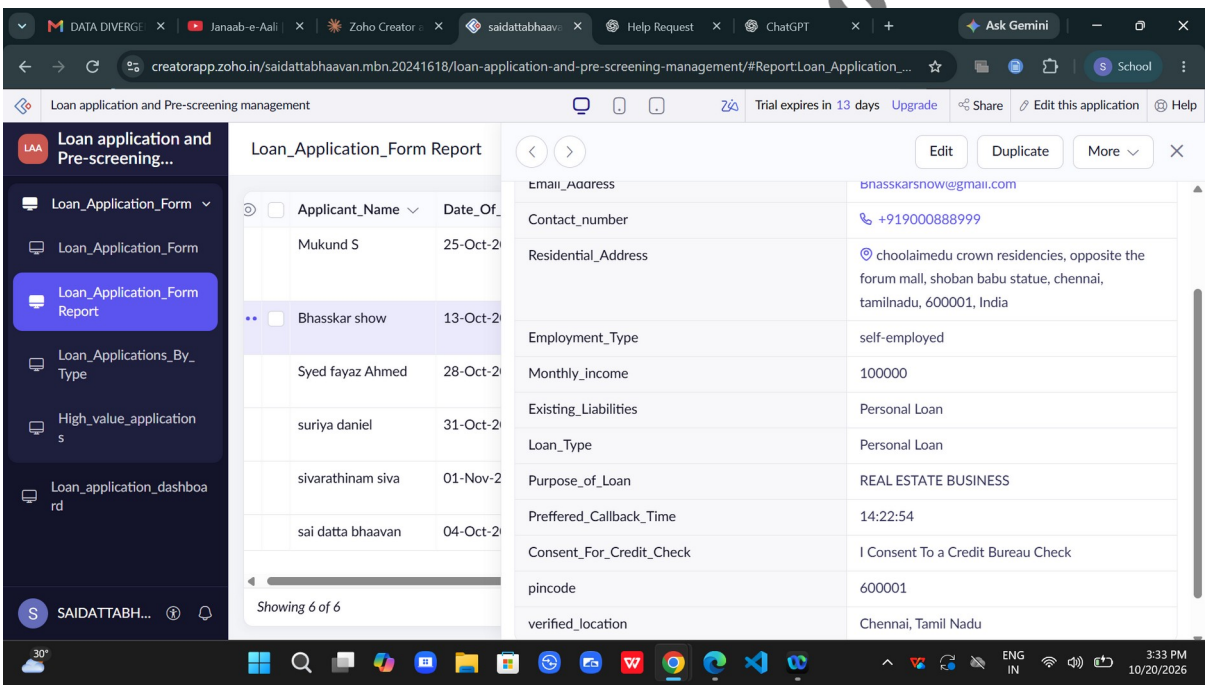


Fig. 8: Report record with PIN 600001 – verified_location filled automatically as "Chennai, Tamil Nadu"

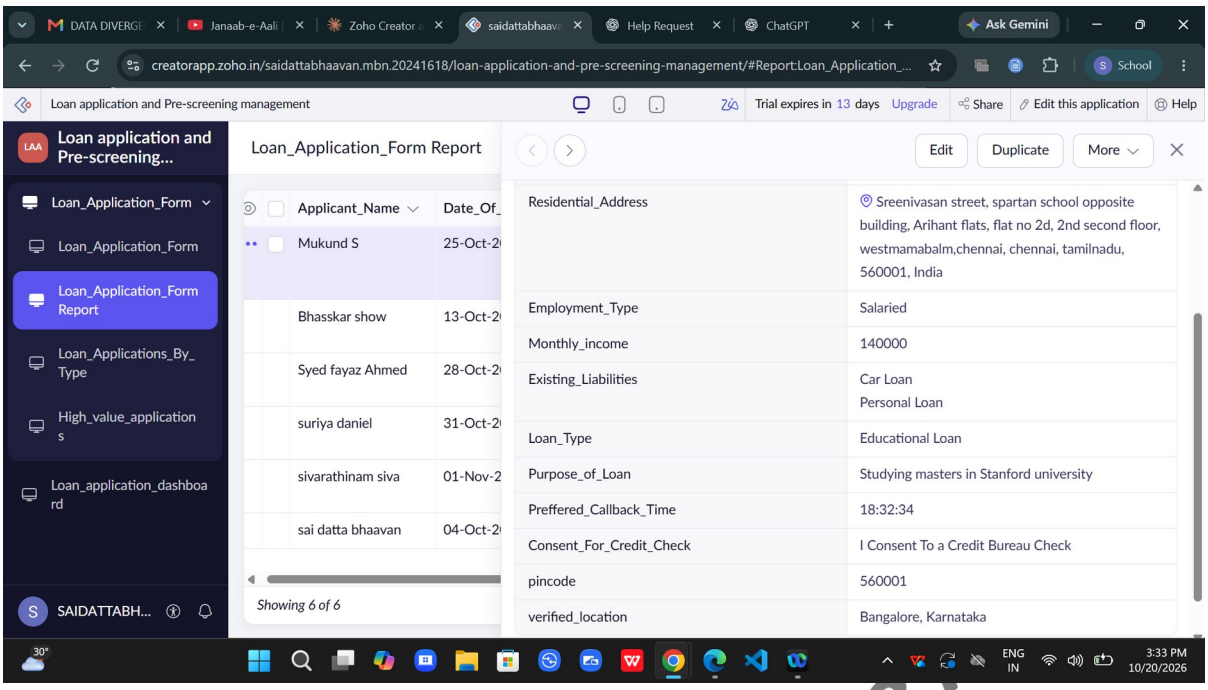


Fig. 9: Report record with PIN 560001 – verified_location filled automatically as "Bangalore, Karnataka"

In Fig. 9 the typed address mentions Chennai while the PIN code 560001 belongs to Bangalore. The automatically verified location exposes this inconsistency, which a loan officer can then check manually.

RESULT

A custom Deluge function was created in Zoho Creator that calls an external REST API using invokeUrl, and it was linked to the Loan Application form through a workflow. On every successful submission the application now fetches the district and state for the entered PIN code and stores them in the verified_location field, as confirmed by the test records. Thus the integration of an external API with a Zoho Creator application was successfully carried out.

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